

Q3LOCK Cubic Graph Embedding and Product-Locality Route Split

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This note records EXP-000796 and strengthens reusable result R-167 to v1.1 without allocating a new result number. It closes the weighted cubic graph-domain obligation left open by EXP-000795, computes the exact heat-simplex coefficient for every prescribed interaction word, and then tests whether this is enough for common real-time dynamics.

It is not enough. Two termwise-absolute organizations fail exactly: raw connected histories have factorial multiplicity, and absolute complex-time continuation has a nonintegrable $e^{C/\varepsilon}$ boundary. These are proof-route no-go results, not nonexistence of the dynamics. The surviving phase-independent target is a first-passage backbone response theorem which resums branches and repeated edges before taking norms. A separate fixed-temperature route requires a uniform fifth onsite-energy moment plus a genuinely two-sided modular or dual-state cutoff estimate.

No common thermodynamic automorphism, common-alpha KMS identification, algebraic ground states, GNS or physical mass gap, continuum removal, physical empty-space comparison, C6, CP1, Sector A or Pre-A closure is claimed. The spatial commutator hierarchy also remains open.

2 Provenance identifiers

- Result: [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#), v1.1, R-167.
- Exploration: [EXP-000796](#).
- Provenance correction: [EXP-000797](#); source-note locator only, with no mathematical change.
- Next gate: [PA-CP1-ST8-Q3LOCK-FIRST-PASSAGE-BACKBONE-REAL-TIME-PRODUCT-AND-ENERGY-TAIL-CLOSURE](#).
- Alternative gate: [PA-CP1-ST8-Q3LOCK-FIFTH-ENERGY-MOMENT-AND-MODULAR-CUTOFF-LOCALITY](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-UNWEIGHTED-MOVING-SITE-CUBIC-GRAPH-UNIFORMITY](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-RAW-ABSOLUTE-CONNECTED-HISTORY-ANIMAL-MAJORANT](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-ABSOLUTE-HEAT-STRIP-REAL-TIME-CONTINUATION](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-DUHAMEL-INNER-PRODUCT-ONLY-COMMON-DYNAMICS](#).

3 Exact weighted cubic graph theorem

For the centered weight $f_x = e^{-\mu d(x,X)}$, write

$$A = 1 + E_f = 1 + T_f + U_f, \quad U_f \geq \gamma f_x |q_x|^4, \quad (3.1)$$

where $0 < \gamma < g/32$ and

$$S_f \leq |X| \left(\frac{1 + e^{-\mu}}{1 - e^{-\mu}} \right)^3. \quad (3.2)$$

The exact R-167 Laplacian input is

$$L_f \leq C_2 \varepsilon U_f + S_f \left(8r_+ + 8cz + \frac{C_2}{4\varepsilon\gamma} \right), \quad C_2 = 3g + 21\lambda. \quad (3.3)$$

On the finite-volume Schwartz core, the exact Schrodinger cross identity expands the full graph norm as

$$\begin{aligned} \|A\psi\|^2 &= \|T_f\psi\|^2 + \|U_f\psi\|^2 + \|\psi\|^2 + 2\langle T_f \rangle + 2\langle U_f \rangle \\ &\quad + \frac{D_f}{\chi} - \frac{\hbar^2}{2\chi} \langle L_f \rangle. \end{aligned} \quad (3.4)$$

Choose

$$\varepsilon_* = \frac{4\chi}{\hbar^2 C_2}, \quad b_* = S_f \left(8r_+ + 8cz + \frac{\hbar^2 C_2^2}{16\chi\gamma} \right), \quad \beta_* = \frac{\hbar^2 b_*}{2\chi}. \quad (3.5)$$

The negative U_f coefficient in (3.4) then cancels exactly. Since $A \geq 1$,

$$\|U_f\psi\|^2 \leq \max(1, \beta_*) \|A\psi\|^2, \quad \|U_f A^{-1}\| \leq \kappa := \sqrt{\max(1, \beta_*)}. \quad (3.6)$$

The finite polynomial Schrodinger operator has the stated Schwartz core as an operator core. Heinz-Kato graph interpolation therefore gives

$$\|U_f^\theta A^{-\theta}\| \leq \kappa^\theta, \quad 0 \leq \theta \leq 1. \quad (3.7)$$

Because U_f and q_x commute as multiplication operators,

$$\boxed{f_x^{m/4} \| |q_x|^m A^{-m/4} \| \leq \gamma^{-m/4} \kappa^{m/4}, \quad 0 \leq m \leq 4.} \quad (3.8)$$

The adjoint orientation has the same bound. In particular,

$$f_x^{3/4} \| |q_x|^3 A^{-3/4} \| \leq \gamma^{-3/4} \max(1, \beta_*)^{3/8}. \quad (3.9)$$

For each Q3 component,

$$|\partial_e W_4(q)| \leq (g + 12\lambda) |q|^3, \quad (3.10)$$

so (3.9) also closes the cubic onsite force. The full eight-component gradient costs an additional factor $\sqrt{8}$.

4 Sharp spatial boundary and moving centers

A compact product bump translated by amplitude R at site x satisfies

$$\|q_{x,e}^3 \psi_{x,R}\| \geq (R-1)^3, \quad \|A\psi_{x,R}\| \leq C_0 + C_1 f_x R^4. \quad (4.1)$$

Hence spectral interpolation forces every unweighted moving-site constant to obey

$$C_x \geq C_1^{-3/4} f_x^{-3/4}. \quad (4.2)$$

This rejects support-location-uniform unweighted bounds, not boundedness at a fixed x .

The energy can instead be recentered. For neighboring centers x, y , let $\vartheta = e^\mu - 1$. The weighted kinetic terms commute, the potentials are nonnegative, and (3.6) gives

$$\|(U_x - U_y)A_y^{-1}\| \leq \vartheta\kappa, \quad \|(T_x - T_y)A_y^{-1}\| \leq \vartheta(2 + \kappa). \quad (4.3)$$

Thus, in both orientations,

$$\|A_x A_y^{-1}\| \leq C_\mu := 1 + 2(e^\mu - 1)(1 + \kappa), \quad \|A_x^s A_y^{-s}\| \leq C_\mu^s, \quad 0 \leq s \leq 1. \quad (4.4)$$

A fixed graph exponent therefore pays only C_μ^{sn} along an n -step backbone.

5 Prescribed-word heat-simplex theorem

Let $K \geq 1$ and suppose every bond obeys

$$\|V_e K^{-1/2}\|, \|K^{-1/2} V_e\| \leq b. \quad (5.1)$$

Using $\|K^{1/2} e^{-sK}\| \leq (2es)^{-1/2}$ and the exact Dirichlet integral,

$$\int_{\sum s_j = \beta} \prod_{j=0}^{n-1} s_j^{-1/2} ds = \frac{\pi^{n/2} \beta^{n/2}}{\Gamma(1 + n/2)}, \quad (5.2)$$

one obtains for every prescribed edge word

$$\boxed{\|\mathcal{I}_{\mathbf{e},n}(\beta)\| \leq \frac{[b\sqrt{\pi/(2e)}\sqrt{\beta}]^n}{\Gamma(1 + n/2)}}. \quad (5.3)$$

For $V_{xy} = -cq_x \cdot q_y$, quartic coercivity gives

$$b = \frac{c}{\sqrt{2\gamma}}, \quad \kappa_0 = c\sqrt{\frac{\pi}{e\gamma}} \quad (5.4)$$

after the two commutator orientations are summed.

6 Why raw absolute histories fail

Take a length- m backbone and attach four distinct transverse leaves at each of m selected vertices. This $5m$ -edge animal admits at least $(4m)!$ connected growth histories. Multiplying the history count by the uniform per-word bound (5.3) gives

$$M_m = \frac{(4m)! a^{5m}}{\Gamma(1 + 5m/2)}, \quad \log M_m = \frac{3}{2}m \log m + O(m). \quad (6.1)$$

The raw absolute majorant has zero radius. This does not prove that the exact series or dynamics diverges; it proves that branches and repeated edges must be resummed before taking norms.

Even granting a reduction to edge chains, the absolute complex-time estimate at $\zeta = \varepsilon + it/\hbar$ contains

$$\exp \left[-\rho d + P^2 \kappa_0^2 e^{2\rho} \frac{|\zeta|^2}{\varepsilon} \right], \quad P = 2z - 1 = 11. \quad (6.2)$$

For $t \neq 0$, every finite Balakrishnan power encounters

$$\int_0^1 \varepsilon^{s-1} e^{C/\varepsilon} d\varepsilon = \infty. \quad (6.3)$$

Thus absolute strip continuation also fails as a proof method. Oscillatory or unitary cancellation remains open.

7 Exact surviving real-time target

Let $\mathcal{R}_{\pi,n}^{(j)}$ sum every branch and repeated-edge history with one first-passage backbone π , before taking a norm. With $s_1 = 1/2, s_2 = 3/4$, the load-bearing target is

$$\int_{\Delta_n(t)} \left(\|\mathcal{R}_{\pi,n}^{(j)} A^{-s_j}\| + \|A^{-s_j} \mathcal{R}_{\pi,n}^{(j)}\| \right) d\mathbf{t} \leq C_j \frac{[G_j \sqrt{|t|/\hbar}]^n}{\Gamma(1+n/2)}. \quad (7.1)$$

It would imply

$$C_j(d, t) \leq \frac{2C_j z |X|}{P} \exp \left[-\rho d + \frac{P^2 G_j^2 e^{2\rho}}{\hbar} |t| \right], \quad (7.2)$$

with velocity

$$v_{\rho,j} = \frac{P^2 G_j^2 e^{2\rho}}{\rho \hbar}. \quad (7.3)$$

The moving-center factor from (4.4) is absorbed into G_j . If $\rho > \mu/4$, the conditional R-167 Duhamel theorem then supplies two-sided volume Cauchy and strong-star core dynamics. Equation (7.1), the branch resummation, and a noncollapsing Hamiltonian algebra remain open.

8 Equilibrium cutoff alternative

At fixed temperature, split

$$H_\Lambda(h) + C|\Lambda| = \sum_x k_{x,h} + \sum_{\langle xy \rangle} V_{xy}, \quad k_{x,h} \geq 1 + \gamma |q_x|^4. \quad (8.1)$$

For $P_{x,R} = 1_{[k_{x,h} \leq R]}$, projected bonds are $O(R^{1/2})$. If one separately proves both a local moment

$$\sup_{\Lambda, h, x} \varphi_{\Lambda, h}(k_{x,h}^p) \leq M_p, \quad p > d + 1, \quad (8.2)$$

and a genuinely two-sided nonracial cutoff estimate, then $R_m = m^b$ with

$$\frac{2d}{p-1} < b < 2 \quad (8.3)$$

balances the leakage and factorial tails. In $d = 3, p = 5, b = 7/4$ give exponents $-1/2$ and $-1/8$ in the two leading asymptotic tests.

The moment alone is not enough. Let

$$He_n = ne_n, \quad p_n = (1 - e^{-\beta})e^{-\beta n}, \quad X_n = |e_n\rangle\langle e_0|. \quad (8.4)$$

Both squared Duhamel norms satisfy

$$(X_n, X_n)_D = (X_n^*, X_n^*)_D = \frac{p_0 - p_n}{\beta n} \longrightarrow 0, \quad (8.5)$$

but $X_n e_0 = e_n$ and the symmetric GNS square norm tends to $p_0/2 > 0$. Hence Duhamel convergence does not imply strong-star dynamics. Uniform modular bandwidth Ω would repair the topology via

$$\|X\|_{\rho, \#}^2 \leq \left[\frac{\beta\Omega}{2} \coth\left(\frac{\beta\Omega}{2}\right) \right] (X, X)_D, \quad (8.6)$$

while Q3LOCK currently requires a high-modular-energy tail theorem instead.

9 Devil’s-advocate review and verification boundary

The primary implementation derives every coefficient and finite check from declared inputs. A non-importing standard-library implementation reconstructs the Q3 graph, graph constant, heat coefficients, animal asymptotics, cutoff exponents and Duhamel fixture independently. The integrated verifier reruns both, compares their derived core, verifies source freshness and requires all formal authorities and this rendered PDF.

Adversarial review confirmed the operator core, Heinz–Kato orientation, $\sqrt{8}$ vector factor, moving-site boundary, integrable simplex endpoints, animal Stirling coefficient, and the method-only scope of both heat obstructions. It also upheld that onsite moments alone cannot justify nonracial strong-star convergence.

10 Reproduction

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_cubic_graph_product_locality_route_split.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_cubic_graph_product_locality_route_split_
independent.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_cubic_graph_product_locality_route_split_
verify.py
```

Run the primary and independent implementations first and the integrated freshness/provenance verifier last.

11 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167 v1.1; no new result number.
Exploration:	EXP-000796.
Precise statement:	Weighted cubic graph embedding, Q3 force multiplier, moving-center graph comparison, and prescribed-word heat simplex proved. Raw absolute animal and strip continuations rejected. First-passage RT product and

fifth-energy/modular cutoff routes remain open.
 Scope: T0 finite-volume analytic theorem, exact route
 counterexamples, and conditional locality reductions.
 Dependencies: EXP-000792--EXP-000797; exact fixed-volume
 ST8/Q3LOCK Hamiltonian and R-167 v1.0 energy bounds.
 Evidence grade: ANALYTIC + EXACT EXECUTED + INDEPENDENT + PDF QA.
 Reproduction command: Run the three commands under Reproduction in the
 listed order, with the integrated verifier last.
 Expected output: Primary PASS 66/66; independent PASS 143/143;
 integrated PASS 273/273.
 Falsification gate: Any graph-expansion sign, Heinz--Kato domain,
 Q3 force, moving-center, heat-simplex, animal,
 strip, Duhamel-topology, hash, scope or provenance
 mismatch.
 Tier before / after: C6 T1 / C6 T1; no claim or tier change.
 No-overclaim statement: First-passage spatial commutator/product closure,
 fifth energy moment, modular tail, common alpha/KMS,
 algebraic ground states,
 GNS gap, continuum, physical empty, C6, CP1,
 Sector A and Pre-A remain open.
 Next required action: Prove either the first-passage RT-PV response plus
 Hamiltonian energy-tail algebra, or the fixed-beta
 fifth-energy and modular/dual cutoff locality gate.